

How Open Source Is Making QUANTUM COMPUTING Accessible To Everyone

Open source initiatives are breaking down the barriers to quantum computing, making it accessible to everyone. Explore what quantum computing is, the challenges of traditional adoption, how open source platforms are democratising the technology, and how you can get started in this exciting domain.



The author, **DHAVAL GAJJAR**, is the CTO of Textdrip, and the CEO of Pranshtech Solutions and WeTechnolabs Solutions. With a passion for technology and innovation, he explores the ever-evolving world of digital solutions, sharing insights and expertise to drive progress in the tech industry.

Imagine a small startup working on optimising supply chains for sustainable farming. Traditionally, they'd need powerful computing resources to model complex systems like weather patterns, crop yields, and transportation logistics—tasks that can take classical computers weeks or even months. Accessing quantum computing could solve these problems in hours, but the cost of quantum hardware and expertise often makes it inaccessible for smaller organisations. This is where open source quantum computing steps in.

Quantum computing is no longer a distant frontier—open source is making

it a shared journey for everyone, unlocking limitless possibilities for innovation and collaboration.

What is quantum computing?

Quantum computing is a new approach to computation that leverages the principles of quantum mechanics, a branch of physics that describes the behaviour of particles at the smallest scales (atoms and subatomic particles). Unlike classical computers, which process information in binary bits (0s or 1s), quantum computers use quantum bits, or qubits, that can exist in multiple states at once. This allows quantum computers to solve certain